

## Lecture 6

# When integrability fails

**Objectives.** Lectures 4 and 5 built the integrable world: tori, actions, frequencies. This lecture perturbs it. Poincaré's theorem says that a perturbation, under conditions we state exactly, destroys the integrals; the KAM theorem says that most of the tori survive anyway; between the two, the resonant tori break into island chains and chaos. We watch all of it happen in the Hénon–Heiles system, put it beside the Toda chain it truncates, and end with the precise sense — and the only sense this course claims — in which integrability is exceptional.

## 6.1 Poincaré's non-existence theorem

Imported from [Lecture 5](#): near a torus of an integrable system there are action–angle coordinates  $(\varphi, I)$ ,  $\varphi \in T^n$ ,  $I$  in an open set  $D \subset \mathbb{R}^n$ , in which  $H = H_0(I)$  and the motion is  $\dot{\varphi} = \nu(I) = \partial H_0 / \partial I$ ,  $\dot{I} = 0$  ([Theorem 5.5](#), [Corollary 5.4](#)).  $H_0$  is non-degenerate if  $\det(\partial^2 H_0 / \partial I^2) \neq 0$ , and then, for  $n \geq 2$ , the resonant tori of [Definition 4.16](#) are dense and of measure zero ([§ 5.5](#)).

Poincaré called the following *the fundamental problem of dynamics*: a nearly integrable system

$$H(\varphi, I) = H_0(I) + \varepsilon H_1(\varphi, I) + O(\varepsilon^2), \quad H_1 = \sum_{k \in \mathbb{Z}^n} h_k(I) e^{ik \cdot \varphi}, \quad (6.1)$$

with  $\varepsilon$  small and  $H_1$  expanded in a Fourier series in the angles. The planets are the original case:  $H_0$  is the Kepler motion of each planet round the Sun, and  $\varepsilon H_1$  their mutual attraction — with the complication that Kepler's  $H_0$  is degenerate ([§ 5.5](#)).

**Example 6.1** (Two rotors in a weak periodic potential). On  $T^2 \times \mathbb{R}^2$ , with the positions  $q \in T^2$  themselves the angles and  $p = I$ , let

$$H = \frac{1}{2} |p|^2 + \varepsilon V(q), \quad H_0 = \frac{1}{2} |I|^2, \quad \nu(I) = I.$$

Here  $\partial^2 H_0 / \partial I^2$  is the identity: non-degenerate. Two choices of  $V$  will do different things.

1.  $V = \cos q_1 + \cos q_2$  separates:  $F = \frac{1}{2} p_1^2 + \varepsilon \cos q_1$  is an exact second integral, and the system is integrable for every  $\varepsilon$ .
2.  $V = P_r(q_1)P_r(q_2)$ , where  $P_r(x) = (1 - r^2)/(1 - 2r \cos x + r^2) = \sum_{m \in \mathbb{Z}} r^{|m|} e^{imx}$  for a fixed  $0 < r < 1$ . Then  $h_k = r^{|k_1| + |k_2|}$ : every Fourier coefficient is nonzero.<sup>1</sup>

What separates the two cases is which Fourier modes act on which resonant tori.

**Definition 6.2** (Poincaré set,  $n = 2$ ). The Poincaré set  $\mathfrak{P} \subset D$  of (6.1) is the set of  $I$  at

<sup>1</sup>`scripts/check-l6-algebra.py`, check 3: the closed form of  $P_r$  symbolically, the coefficients sampled.

which some  $k \in \mathbb{Z}^2 \setminus \{0\}$  has

$$k \cdot \nu(I) = 0 \quad \text{and} \quad h_k(I) \neq 0 :$$

the resonant tori on which a Fourier mode of the perturbation resonates.

For general  $n$ ,  $I \in \mathfrak{P}$  asks for  $n - 1$  linearly independent such  $k$ . In [Example 6.1](#),  $k \cdot \nu(I) = k \cdot I = 0$  is a line through the origin of the  $I$ -plane. With  $V = \cos q_1 + \cos q_2$  only  $k = \pm(1, 0), \pm(0, 1)$  have  $h_k \neq 0$ , and  $\mathfrak{P}$  is the two axes. With  $V = P_r P_r$  every rational line through 0 is in  $\mathfrak{P}$ : it is dense.

**Theorem 6.3** (Poincaré; in Kozlov's form · cited). *Let  $H(\varphi, I, \varepsilon)$  be real-analytic on  $T^n \times D \times (-\varepsilon_0, \varepsilon_0)$ , with  $H(\varphi, I, 0) = H_0(I)$  non-degenerate on  $D$ , and let the Poincaré set  $\mathfrak{P}$  be dense in  $D$ . Then Hamilton's equations have no first integral*

$$F = F_0(\varphi, I) + \varepsilon F_1(\varphi, I) + \varepsilon^2 F_2(\varphi, I) + \cdots ,$$

*a formal power series in  $\varepsilon$  with smooth coefficients, that is independent of  $H$ .*

Here *independent* is [Definition 4.1](#) read order by order:  $F$  and  $H$  are dependent when every  $2 \times 2$  minor of the Jacobian of  $(H, F)$  vanishes identically as a series in  $\varepsilon$  (Kozlov, ch. IV §1, Theorem 2). So the second choice of  $V$  in [Example 6.1](#) has no second integral of this kind, and “non-integrable” is a theorem about it, not a report that nobody has found one.

Read the hypotheses as limits on what is claimed.

- Only power series in  $\varepsilon$  with coefficients defined on all of  $T^n \times D$ . Invariant sets that are not level sets of such an  $F$  are not excluded — and § 6.3 will find a great many of them.
- Only non-degenerate  $H_0$ . The isotropic oscillator, with  $H_0 = \nu(I_1 + I_2)$ , is degenerate, and the theorem says nothing about its perturbations. Hénon–Heiles (§ 6.5) is one.
- Only a dense  $\mathfrak{P}$ . With  $V = \cos q_1 + \cos q_2$  it fails, and so does the conclusion.

The engine of the proof is visible at first order, and we prove that part in the next section.

## 6.2 Small denominators

The obvious strategy is to remove  $\varepsilon H_1$  by a canonical transformation close to the identity, as Hamilton–Jacobi removed all of  $H$  in [Lecture 5](#). Take a type-2 generating function ([Proposition 3.16](#)) with old positions  $\varphi$  and new momenta  $I'$ ,

$$S(\varphi, I') = I' \cdot \varphi + \varepsilon S_1(\varphi, I'), \quad I = \frac{\partial S}{\partial \varphi} = I' + \varepsilon \frac{\partial S_1}{\partial \varphi}, \quad \varphi' = \frac{\partial S}{\partial I'} = \varphi + \varepsilon \frac{\partial S_1}{\partial I'}.$$

Substitute into (6.1) and expand  $H_0(I' + \varepsilon \partial_\varphi S_1)$ :

$$H = H_0(I') + \varepsilon \left( \nu(I') \cdot \frac{\partial S_1}{\partial \varphi} + H_1(\varphi, I') \right) + O(\varepsilon^2). \quad (6.2)$$

The new Hamiltonian is integrable to first order if the bracket does not depend on the angles. Its angle average is  $h_0$ , since  $\partial_\varphi S_1$  averages to zero; so we need the *homological equation*

$$\nu(I') \cdot \frac{\partial S_1}{\partial \varphi} = h_0(I') - H_1(\varphi, I'). \quad (6.3)$$

**Proposition 6.4** (First-order perturbation theory). *On an open set of  $I'$  on which  $k \cdot \nu(I') \neq 0$  for every  $k$  with  $h_k \neq 0$ , and on which the series below converges together with its first and second derivatives, and for  $\varepsilon$  small enough,*

$$S_1(\varphi, I') = \sum_{k \neq 0} \frac{i h_k(I')}{k \cdot \nu(I')} e^{ik \cdot \varphi} \quad (6.4)$$

(terms with  $h_k = 0$  omitted) solves (6.3), and the new Hamiltonian is  $H_0(I') + \varepsilon h_0(I') + O(\varepsilon^2)$ .

*Proof.*  $\nu \cdot \partial_\varphi e^{ik \cdot \varphi} = i(k \cdot \nu) e^{ik \cdot \varphi}$ , so the  $k$ -th mode of the left side of (6.3) is  $i(k \cdot \nu)s_k$  and that of the right side is  $-h_k$ . Solve:  $s_k = ih_k/(k \cdot \nu)$ .<sup>2</sup> ■

If  $H_1$  has finitely many Fourier modes, such open sets exist: everything off the finitely many active resonance hypersurfaces  $k \cdot \nu(I') = 0$ . If the Poincaré set is dense, no open set qualifies.

Every term is divided by  $k \cdot \nu(I')$ , the combination whose vanishing is resonance. On a resonant torus with  $h_k \neq 0$  — a point of the Poincaré set — the  $k$ -th mode of (6.3) reads  $0 = -h_k$ , and there is no solution at all. Near one the coefficient is enormous. And for  $n \geq 2$  the resonant tori are dense, so every torus has resonant ones arbitrarily close: these are the *small denominators* (Figure 6.1). They are not a defect of the method. Near resonance, a small perturbation is not a small change.

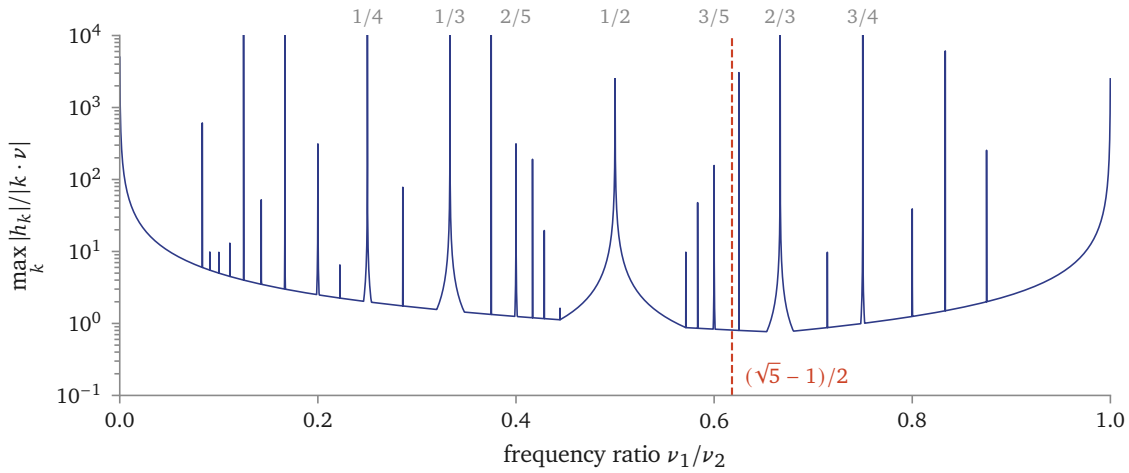


Figure 6.1: Small denominators for the rotors of Example 6.1 with  $V = P_r P_r$ ,  $r = \frac{1}{2}$ : the largest term  $|h_k|/|k \cdot \nu|$  of (6.4), over  $0 < |k_1| + |k_2| \leq 40$ , against the frequency ratio. Each rational  $a/b$  with  $a + b \leq 40$  is a spike to infinity (drawn truncated); as the cut-off grows the spikes become dense, but their widths shrink like  $r^{a+b}$ , and most of the line stays low. The golden ratio sits as far from them as a number can (scripts/figures/small\_denominators.py).

The same obstruction kills integrals, which is Poincaré's point.

**Proposition 6.5** (Poincaré's mechanism at first order). *Let  $n = 2$ , let  $H_0$  be non-degenerate, and let  $F = F_0 + \varepsilon F_1 + O(\varepsilon^2)$  be a first integral of (6.1), with  $F_0, F_1$  smooth on  $T^2 \times D$ . Then*

1.  $F_0$  depends on  $I$  alone;

<sup>2</sup>scripts/check-l6-algebra.py, check 1, symbolically, including the sign through the type-2 convention of Proposition 3.16.

2. at every point of  $\mathfrak{P}$ ,  $\nabla F_0$  is parallel to  $\nu$ , that is,  $dF_0 \wedge dH_0 = 0$ .

If  $\mathfrak{P}$  is dense,  $dF_0 \wedge dH_0 = 0$  everywhere: the leading term of every such integral is a function of the energy.

*Proof.* Expand  $\{F, H\} = 0$  in  $\varepsilon$ , with the angles as positions and the actions as momenta.

Order 1:  $\{F_0, H_0\} = \nu \cdot \partial_\varphi F_0 = 0$ . Its  $k$ -th Fourier mode is  $i(k \cdot \nu) f_{0,k}(I) = 0$ , so  $f_{0,k}(I) = 0$  wherever  $\nu(I)$  is non-resonant. Those  $I$  are dense, because  $H_0$  is non-degenerate (§ 5.5), and  $f_{0,k}$  is continuous, so  $f_{0,k} \equiv 0$  for  $k \neq 0$ :  $F_0 = F_0(I)$ .

Order  $\varepsilon$ :  $\{F_1, H_0\} + \{F_0, H_1\} = 0$ . The first term is  $\nu \cdot \partial_\varphi F_1$ ; the second is  $-\nabla F_0 \cdot \partial_\varphi H_1$ , since  $F_0$  has no angles. The  $k$ -th mode:<sup>3</sup>

$$i(k \cdot \nu) f_{1,k} = i(k \cdot \nabla F_0) h_k.$$

At  $I \in \mathfrak{P}$  pick the  $k$  of Definition 6.2: the left side vanishes and  $h_k \neq 0$ , so  $k \cdot \nabla F_0 = 0$ . Then  $\nu$  and  $\nabla F_0$  are both orthogonal to the nonzero vector  $k$  of  $\mathbb{R}^2$ , hence parallel. The last sentence follows by continuity. ■

Where  $dH_0 \neq 0$ , the conclusion makes  $F_0$  constant along the level curves of  $H_0$ , so locally a function of it. This is not yet non-existence: the rest needs all orders.

#### Reading

First order alone cannot give more:  $F = H + \varepsilon G(I)$ , for any  $G$ , satisfies  $\{F, H\} = O(\varepsilon^2)$  and has  $F_0 = H_0$ . Kozlov's proof of Theorem 6.3 subtracts that function of  $H$  from  $F$ , divides by  $\varepsilon$ , and repeats the argument on the new series, order by order. For general  $n$  the same computation at a point of  $\mathfrak{P}$  puts  $\nabla F_0$  orthogonal to  $n - 1$  independent vectors  $k$ , all orthogonal to  $\nu$  as well, and so again parallel to  $\nu$ ; that is why the general Poincaré set asks for  $n - 1$  of them.

## 6.3 The KAM theorem

Poincaré's theorem destroys the integrals, but not necessarily the tori. A torus is an invariant set, and it may survive even where no integral does; Proposition 6.4 cannot build it, because of the resonances. The tori to hope for are those *furthest* from every resonance.

**Definition 6.6 (Diophantine).** With  $|k| = |k_1| + \dots + |k_n|$ , a frequency vector  $\nu \in \mathbb{R}^n$  is  $(\gamma, \tau)$ -Diophantine if

$$|k \cdot \nu| \geq \frac{\gamma}{|k|^\tau} \quad \text{for every } k \in \mathbb{Z}^n \setminus \{0\}.$$

It is quantitative non-resonance:  $k \cdot \nu$  may get small as  $|k|$  grows, but only polynomially. Then every term of (6.4) is at most  $|h_k| |k|^\tau / \gamma$ , which an analytic  $H_1$ , with  $h_k$  decaying exponentially in  $|k|$ , easily beats. The golden vector  $\nu = (\frac{1}{2}(\sqrt{5} - 1), 1)$  is Diophantine with  $\tau = 1$  (Sheet 6). And almost every vector is Diophantine.

**Lemma 6.7 (Diophantine vectors have almost full measure,  $n = 2$ ).** Let  $\tau > 1$ . In the disc  $|\nu| \leq R$  of  $\mathbb{R}^2$ , the vectors that are not  $(\gamma, \tau)$ -Diophantine have total area at most  $16\sqrt{2} R \gamma \zeta(\tau)$ , where  $\zeta(\tau) = \sum_{m \geq 1} m^{-\tau}$ .

<sup>3</sup>`scripts/check-l6-algebra.py`, check 2, with the frozen bracket.

*Proof (★).* For one  $k \neq 0$ , the bad set  $|k \cdot \nu| < \gamma |k|^{-\tau}$  is a strip about the line  $k \cdot \nu = 0$  of width  $2\gamma |k|^{-\tau} / \|k\| \leq 2\sqrt{2} \gamma |k|^{-\tau-1}$ , since the Euclidean length  $\|k\| \geq |k| / \sqrt{2}$ . Inside the disc it has length at most  $2R$ , so area at most  $4\sqrt{2} R \gamma |k|^{-\tau-1}$ . There are  $4m$  vectors  $k$  with  $|k| = m$ . Summing,  $\sum_{m \geq 1} 4m \cdot 4\sqrt{2} R \gamma m^{-\tau-1} = 16\sqrt{2} R \gamma \zeta(\tau)$ , which is finite because  $\tau > 1$ . ■

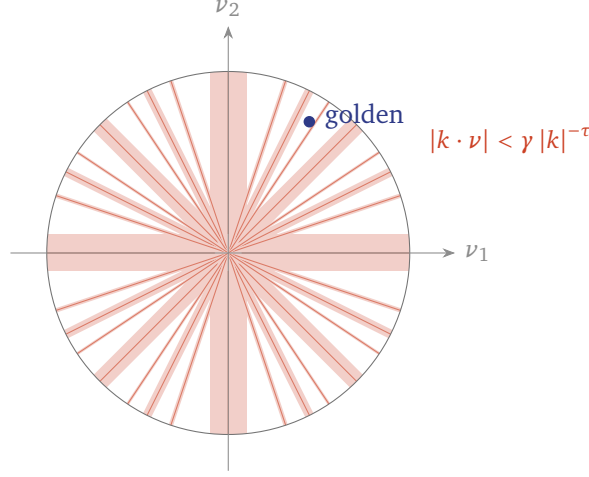


Figure 6.2: **Lemma 6.7.** Each resonance  $k \cdot \nu = 0$  (a line through 0) is thickened to a strip of width  $\propto \gamma |k|^{-\tau-1}$ ; representative strips with  $|k| \leq 5$  are drawn, not to scale. Every line through 0 with rational slope gets a strip, so the strips are dense, but their total area is  $O(\gamma)$ : the Diophantine vectors, which lie outside all of them, are almost all.

**Theorem 6.8** (Kolmogorov–Arnold–Moser · cited). *Let  $H = H_0(I) + \varepsilon H_1(\varphi, I)$  on  $T^n \times D$ , with  $H_0$  analytic and  $H_1$  analytic and  $2\pi$ -periodic in the angles. Let  $D_0$  be a bounded region whose closure lies in  $D$ , with  $H_0$  non-degenerate on a neighbourhood of that closure, and let both functions extend analytically to a complex neighbourhood of fixed width of  $T^n \times \overline{D_0}$ , where  $H_1$  is bounded. Then for sufficiently small  $\varepsilon$ , most non-resonant invariant tori of  $H_0$  over  $D_0$  do not vanish but are only slightly deformed: the perturbed system has invariant tori, densely filled by quasi-periodic orbits with  $n$  independent frequencies. They form a majority in the sense that the measure of the part of  $T^n \times D_0$  they do not cover is small when  $\varepsilon$  is small. The theorem guarantees the survival of a large family whose frequencies satisfy a fixed Diophantine condition (Definition 6.6).*

That is Arnold’s statement (App. 8 B, *Kolmogorov’s theorem*, and the hypotheses on the following page); the last sentence is from his sketch of the proof, which keeps the tori with  $|k \cdot \nu| \geq C |k|^{-\tau}$  and discards the rest — a set of small measure when  $C$  is small, as Lemma 6.7 shows. Three things the theorem does not say. It gives no size for “sufficiently small” and no rate for “small”. It says nothing about the resonant tori: that is the next section. And it gives no integral: the surviving tori are indexed by Diophantine frequencies, a set with no interior, and leave gaps round every resonance, which by Theorem 6.3 no integral that is a power series in  $\varepsilon$  fills in. That is how the two theorems are both true. The proof builds each surviving torus by Newton’s method rather than by the series (6.4), and uses Definition 6.6 at every step to control the denominators; we do not give it.

For  $n = 2$  it has a sharp consequence. On a three-dimensional energy level a two-dimensional torus separates, like a surface of revolution in space: an orbit starting between

two surviving tori stays between them forever, and its actions stay near their initial values (Arnold, App. 8, p. 407, under the analogous *isoenergetic* non-degeneracy condition). For  $n \geq 3$  the tori separate nothing, and a slow drift through the gaps is possible — *Arnold diffusion*, which we only name.

## 6.4 Poincaré–Birkhoff and island chains

To see what becomes of the resonant tori we need one tool, which is also the tool of the next section: the section map.

### The section map

Take  $n = 2$  and fix the energy,  $H = E$ , a three-dimensional level set. Choose a surface  $\Sigma$  in it that the flow crosses transversally — for instance  $q_1 = 0$  with  $\dot{q}_1 > 0$  — with coordinates  $(q_2, p_2)$ , the remaining momentum  $p_1$  being fixed by  $H = E$ . If the orbit from  $x \in \Sigma$  comes back to  $\Sigma$ , call the first return  $P(x)$ : the *Poincaré map* (Figure 6.3).

**Proposition 6.9** (The section map preserves area · proved modulo Stokes). *For every loop  $\gamma$  in  $\Sigma$  on which  $P$  is defined, the area enclosed in the  $(q_2, p_2)$ -plane is the same for  $\gamma$  and  $P \circ \gamma$ .*

The proof is Stokes on the tube of orbits joining  $\gamma$  to  $P \circ \gamma$ , on which  $\omega$  vanishes; it is in the reading.

*Proof (★).* With  $\tau(u)$  the return time of  $\gamma(u)$ , the map  $(s, u) \mapsto \Phi_{s\tau(u)}(\gamma(u))$ ,  $s \in [0, 1]$ , sweeps a tube  $C$  of orbits from  $\gamma$  to  $P \circ \gamma$ . Both of its tangent vectors are tangent to the level set  $H = E$ , and one of them is  $\tau X_H$ . Since  $\iota_{X_H} \omega = dH$  (Definition 3.5),  $\omega(X_H, v) = dH(v) = 0$  for every  $v$  tangent to the level set, so  $\omega$  vanishes on  $C$ . By Stokes on  $C$  (Theorem 2.23) and  $d(p \cdot dq) = -\omega$  (Theorem 3.28),  $\oint_{P \circ \gamma} p \cdot dq = \oint_{\gamma} p \cdot dq$ . On  $\Sigma$ ,  $dq_1 = 0$ , so  $p \cdot dq = p_2 dq_2$  there, and  $\oint p_2 dq_2$  is the signed enclosed area. So  $P$  preserves area, and orientation with it. ■

This is Theorem 3.19 with a variable time: the invariant is carried from one crossing to the next. Two readings follow. An invariant torus meets  $\Sigma$  in a closed curve, and  $P$  maps the curve to itself. And for the unperturbed system, with  $\Sigma = \{\varphi_2 = 0\}$ , coordinates  $(\varphi_1, I_1)$  and  $\nu_2 > 0$  (reverse  $\varphi_2$  if necessary), the return time is  $2\pi/\nu_2$ , in which  $\varphi_1$  advances by  $2\pi\nu_1/\nu_2$ :

$$P_0(\varphi_1, I_1) = (\varphi_1 + 2\pi\rho(I_1), I_1), \quad \rho = \nu_1/\nu_2 \text{ on the level } H_0 = E. \quad (6.5)$$

Each circle  $I_1 = c$  is turned rigidly by  $2\pi\rho$ . If  $\rho$  changes with  $I_1$  — which is what isoenergetic non-degeneracy says —  $P_0$  is a *twist map*: circles at different heights turn by different angles.

On a circle with  $\rho = a/b$  rational (in lowest terms), every point is a fixed point of  $P_0^b$ : the torus is resonant, filled with periodic orbits (Proposition 4.17). On an irrational circle every orbit is dense. Perturb:  $P_\epsilon$  is still area-preserving (Proposition 6.9) and close to  $P_0$ . By Theorem 6.8 most of the irrational circles survive as invariant curves. A circle of fixed points is not robust — nothing forces a whole circle of solutions of  $P^b x = x$  to persist — and what remains of it is fixed by a theorem of Poincaré.

**Theorem 6.10** (Poincaré’s geometric theorem · cited). *An area-preserving homeomorphism of a closed annulus to itself that turns its two boundary circles in opposite directions has at least two fixed points.*

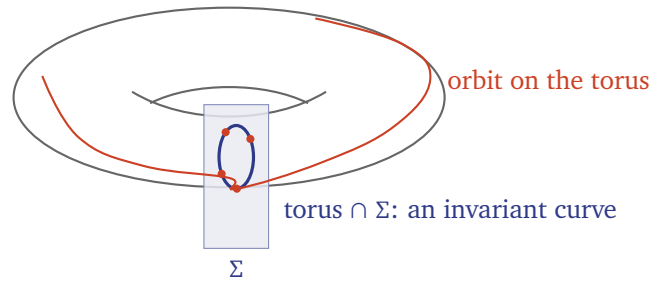


Figure 6.3: A Poincaré section. The orbit (red) winds round an invariant torus and pierces the surface  $\Sigma$  once per turn; the crossings (dots) all lie on the closed curve in which the torus meets  $\Sigma$ , and the return map  $P$  carries the curve to itself.

(Arnold, App. 9: Poincaré announced it just before his death, and Birkhoff proved it.) Apply it between two surviving invariant curves whose rotation numbers  $\rho_- < a/b < \rho_+$  straddle the resonance, to  $P_\epsilon^b$  followed by a rotation by  $-2\pi a$ : the inner boundary turns back, the outer forward. The result is periodic orbits of period  $b$  that survive the break-up — for a sufficiently small perturbation of an isoenergetically non-degenerate system, at least two geometrically distinct ones (Arnold, App. 9, p. 422).

#### Reading

Two steps hide in “the inner boundary turns back”. Arnold’s hypothesis is pointwise: every point of the inner circle must move backwards. That holds because a circle homeomorphism with rotation number  $\rho_- < a/b$  moves *every* point by less than  $2\pi a$  in  $b$  steps (if some point moved by exactly  $2\pi a$  the rotation number would be  $a/b$ ). And the two fixed points of the theorem might lie on one orbit; that there are two distinct periodic orbits *coming from this resonant torus* is Arnold’s corollary (p. 422): when an  $n$ -dimensional torus filled with closed trajectories of an isoenergetically non-degenerate system breaks up under a sufficiently small perturbation, at least  $2^{n-1}$  closed trajectories form, counted with multiplicity, of which at least  $n$  are geometrically distinct.

In the generic case there are  $kb$  periodic points of each of two kinds, and they alternate round the old circle (Ott, §7.2.2): *elliptic* points, surrounded by small invariant curves of their own, and *hyperbolic* points, saddles of  $P_\epsilon^b$ . The resonant torus has become an *island chain* (Figure 6.4).

### A model twist map, and a homoclinic tangle

The simplest area-preserving perturbation of a twist map is the *standard map* on the cylinder,

$$p' = p + K \sin x, \quad x' = x + p' \quad (x \bmod 2\pi), \quad (6.6)$$

the section map of a rotor kicked once per unit time by the potential  $K \cos x$ . Its Jacobian has determinant 1, and at  $K = 0$  it is (6.5) with rotation  $p$ .<sup>4</sup> Figure 6.5 shows the whole story: at  $K = 0.5$  the golden circle survives, deformed, while the resonant circle  $\rho = 1/3$  has broken into three islands and three hyperbolic points, exactly as Figure 6.4 predicts.

The hyperbolic points carry the chaos. A hyperbolic fixed point has a *stable* curve  $W^s$  of points that approach it, and an *unstable* curve  $W^u$  of points that approach it backwards in

<sup>4</sup>`scripts/check-l6-algebra.py`, check 8. This is Chirikov’s form; Ott (7.15) kicks after the turn, a conjugate map.

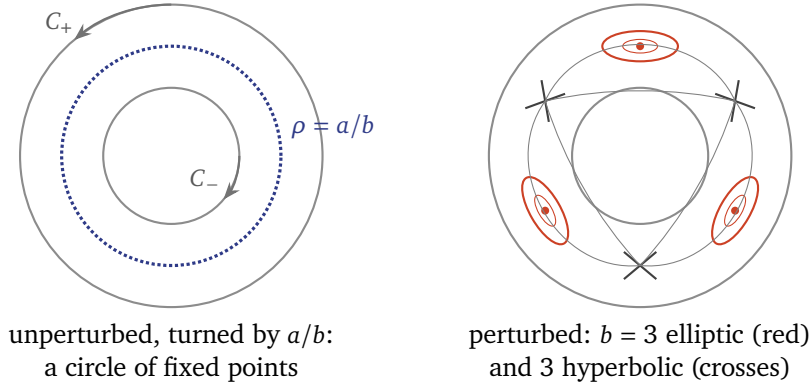


Figure 6.4: Poincaré–Birkhoff. Left: in the frame turning with the resonant circle ( $\rho = a/b$ , dotted), the inner circle  $C_-$  turns backwards and the outer  $C_+$  forwards under  $P_0^b$ ; the dotted circle is fixed pointwise. Right: after the perturbation  $C_\pm$  survive (KAM) and, of the circle, only the fixed points guaranteed by [Theorem 6.10](#) remain — in the generic case alternately elliptic, ringed by islands, and hyperbolic. Schematic; [Figure 6.5](#) is the computed version.

time. In an integrable system they join smoothly into a separatrix, as for the pendulum (§ 5.3). Perturbed, they generically cross. One crossing  $x$  forces infinitely many, since every image  $P^m x$  is again on both curves; the lobes between successive crossings have equal areas because the map preserves area, and as they crowd towards the fixed point they must stretch into long thin tongues. This *homoclinic tangle* ([Figure 6.6](#)), discovered by Poincaré, forces chaotic dynamics nearby (Ott, §§4.3, 7.2.2), and it is where the chaotic sea of [Figure 6.5](#) is born.

## 6.5 Poincaré sections: Hénon–Heiles

Hénon and Heiles wanted to know whether a star in an axisymmetric galaxy has a third integral besides the energy and the angular momentum. The question reduces to two degrees of freedom, and they took as the simplest model

$$H = \frac{1}{2}(p_x^2 + p_y^2) + \frac{1}{2}(x^2 + y^2) + x^2 y - \frac{1}{3}y^3. \quad (6.7)$$

The potential has a minimum at 0 and three saddles, at  $(0, 1)$  and  $(\pm\sqrt{3}/2, -\frac{1}{2})$ , all at height  $1/6$ ; below  $E = 1/6$  the motion near the origin is confined to a curved triangle.<sup>5</sup>

*The energy is the perturbation parameter.* Rescaling  $(x, y, p) \mapsto s(x, y, p)$  gives

$$\frac{1}{s^2}H(sx, sy, sp) = \frac{1}{2}(p_x^2 + p_y^2 + x^2 + y^2) + s(x^2 y - \frac{1}{3}y^3),$$

so the dynamics at energy  $E$  is that of the isotropic oscillator plus a cubic perturbation of size  $\varepsilon = s = \sqrt{E}$  at energy 1. The oscillator is degenerate ( $H_0 = I_1 + I_2$ ), so neither [Theorem 6.3](#) nor [Theorem 6.8](#), as stated, applies. We let the numerics speak.

### How to make a section

Take  $\Sigma = \{x = 0, p_x > 0\}$  at fixed  $E$ , with coordinates  $(y, p_y)$ ; the allowed region is  $\frac{1}{2}p_y^2 + \frac{1}{2}y^2 - \frac{1}{3}y^3 \leq E$ . Launch an orbit from a point of it, with  $p_x = \sqrt{2E - p_y^2 - y^2 + \frac{2}{3}y^3}$ ;

<sup>5</sup>`scripts/check-l6-algebra.py`, check 5.

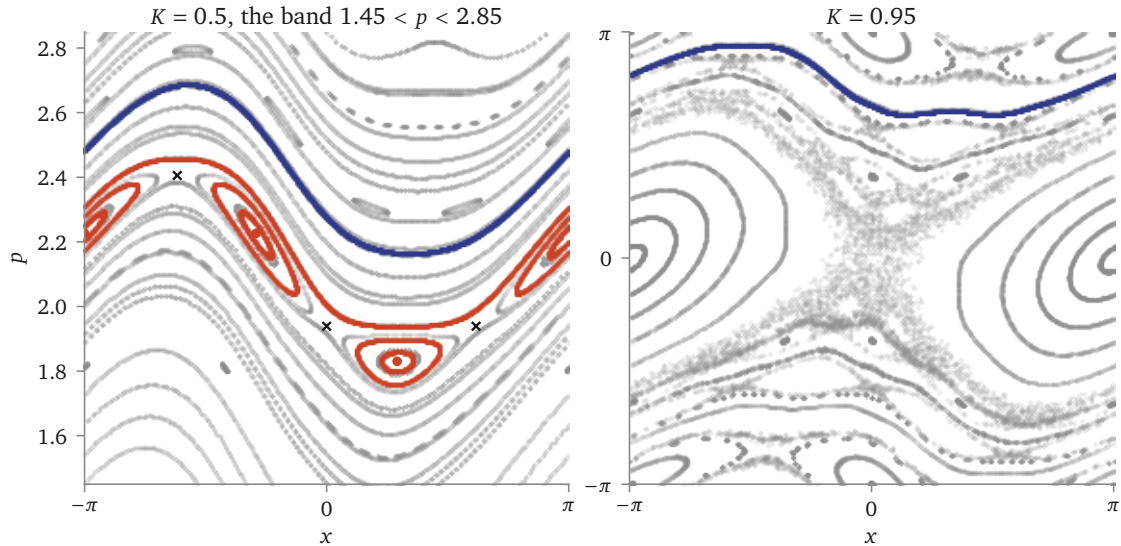


Figure 6.5: The standard map (6.6). Left,  $K = 0.5$ , a band of  $p$ : most circles survive, deformed (grey); the invariant curve of golden rotation number  $(3 - \sqrt{5})/2$  (indigo) is a KAM curve; the resonant circle  $p = 1/3$  is gone, replaced by three islands round an elliptic orbit of period 3 (red) and a hyperbolic orbit of period 3 (crosses). Right,  $K = 0.95$ : a chaotic sea, and the golden curve still there — it is the last to go, at  $K \approx 0.97$  (Greene; Ott, §7.3.2).

integrate Hamilton's equations accurately; record  $(y, p_y)$  each time  $x$  passes through 0 upwards; plot; repeat for a dozen starting points. By Proposition 6.9 the picture is a portrait of an area-preserving map, and the three kinds of orbit look different:

- an orbit on an invariant torus fills a *smooth closed curve*;
- an orbit in an island chain fills several small curves, visited in turn;
- a chaotic orbit fills a *region*, erratically, and its points scatter.

Figure 6.7 is the result, and it is what Hénon and Heiles saw in 1964. At low energy the section is foliated by curves: every orbit looks as if it had a second integral. As  $E$  grows the island chains widen, the tangles between their hyperbolic points thicken into a chaotic layer, and at  $E = 1/6$  the layers have merged into a sea. The *fraction* of the section still covered by curves can be measured (Figure 6.8): it is 1 to the resolution at small  $E$  and collapses near  $E \approx 0.1$ . That resembles the picture Theorem 6.8 gives for non-degenerate systems as  $\varepsilon \rightarrow 0$ , but it is not a prediction of the theorem, which does not cover this degenerate case; it is a measurement.

#### Tutorial

#### The code

The whole computation is a dozen lines of Python.

```
def rhs(t, z):
    x, y, px, py = z
    return [px, py, -(x + 2*x*y), -(y + x*x - y*y)]
def crossing(t, z): return z[0]
crossing.direction = 1
```

# Hamilton's equations for (eq:6:HH)  
# the section  $x = 0$  ...  
# ... crossed upwards,  $px > 0$

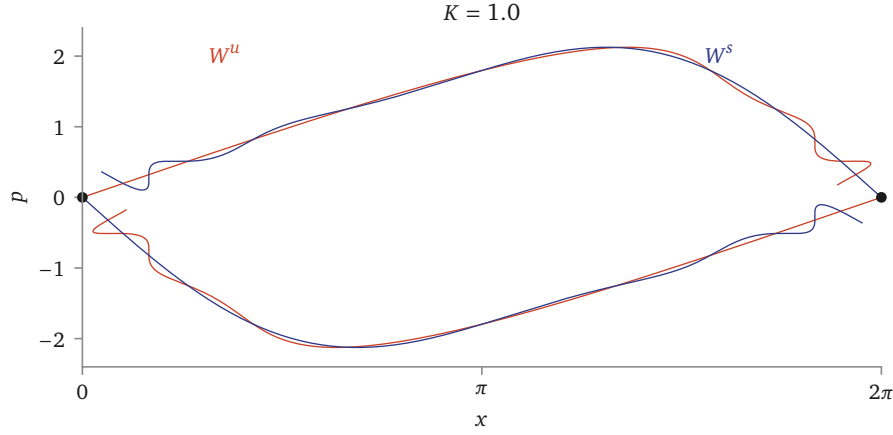


Figure 6.6: A homoclinic tangle, computed: the unstable (red) and stable (indigo) curves of the hyperbolic fixed point of the standard map at  $K = 1$ , which sits at both ends ( $x = 0$  and  $x = 2\pi$  are the same point). They do not join into one separatrix; they cross, and the lobes pile up near the fixed point (`scripts/figures/homoclinic_tangle.py`).

```
def section(E, y, py, tmax=2000):
    px = sqrt(2*E - py**2 - y**2 + 2*y**3/3)
    s = solve_ivp(rhs, (0, tmax), [0, y, px, py], events=crossing,
                  rtol=1e-10, atol=1e-12)
    return s.y_events[0][:, 1], s.y_events[0][:, 3] # (y, py)
```

Two things decide whether the picture can be trusted. The accuracy:  $H$  must stay constant along the orbit, and checking it is part of the computation — the generator measures a drift below  $10^{-8}$  over  $t \approx 2000$ , and refuses to draw above  $5 \times 10^{-8}$ . And the event detection: the crossing must be located by root finding on  $x(t) = 0$ , not by the nearest time step. `scripts/figures/henon_heiles_sections.py` is this code with a loop over launch points.

## 6.6 Toda against Hénon–Heiles at equal energy

Three particles on a ring, each pushed by its neighbours through the exponential potential of Toda, with the centre of mass removed and the coordinates, time and energy rescaled (Toda, §2.9), have the Hamiltonian

$$H_{\text{Toda}} = \frac{1}{2}(p_x^2 + p_y^2) + \frac{1}{24} \left( e^{2y+2\sqrt{3}x} + e^{2y-2\sqrt{3}x} + e^{-4y} \right) - \frac{1}{8}. \quad (6.8)$$

This is the periodic Toda chain, a model that [Lecture 10](#) will prove integrable; here we claim nothing else about it. Expand the exponentials: the constants cancel the  $-\frac{1}{8}$ , the linear terms cancel ( $2y + 2y - 4y = 0$ ), the quadratic ones give  $\frac{1}{2}(x^2 + y^2)$ , and the cubic ones give

$$\frac{1}{144} \left[ (2y + 2\sqrt{3}x)^3 + (2y - 2\sqrt{3}x)^3 - 64y^3 \right] = x^2y - \frac{1}{3}y^3.$$

So  $H_{\text{Toda}}$  agrees with Hénon–Heiles (6.7) through cubic order, and differs from it at quartic order.<sup>6</sup> *Hénon–Heiles is the cubic truncation of the Toda chain.*

<sup>6</sup>`scripts/check-l6-algebra.py`, check 6, which also checks the reduction from three particles.

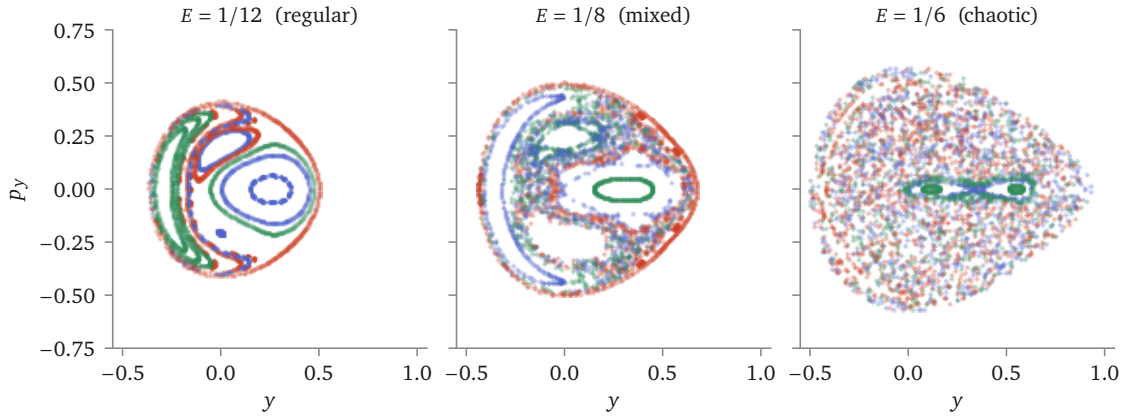


Figure 6.7: Sections  $x = 0$ ,  $p_x > 0$  of Hénon–Heiles (6.7) at three energies, fourteen orbits each; the colours only separate neighbouring orbits. At  $E = 1/12$  most plotted orbits lie on curves and island chains, with a thin chaotic layer already visible. At  $E = 1/8$  curves and islands coexist with a chaotic sea. At  $E = 1/6$  the chaotic orbits fill most of the section; islands remain (scripts/figures/henon\_heiles\_sections.py).

Figure 6.9 puts the two side by side, at the energy at which Hénon–Heiles is most chaotic. Two Hamiltonians that agree through cubic order disagree about everything that matters. The integrable one is not the “simplified” one: dropping terms made the problem harder, not easier. Integrability is a property of the whole Hamiltonian: this example shows that the cubic Taylor polynomial cannot see it, and that it need not survive truncation. That is why this course spends its effort on the exceptional systems exactly as they are, and treats the rest of the world as a perturbation of them — the argument of § 0.5, now with a picture.

## 6.7 Lyapunov exponents, and genericity stated precisely

The eye separates curves from scatter; a number does it better.

**Definition 6.11** (Lyapunov exponent). For an orbit  $\Phi_t(x_0)$  and a tangent vector  $u$  at  $x_0$ ,

$$\lambda(x_0, u) = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|\mathrm{d}\Phi_t(u)|}{|u|},$$

where it exists (Ott, §4.4). A positive exponent means exponential sensitivity to the initial condition.

A positive exponent alone is not chaos: the hyperbolic fixed point of the standard map has one. For a bounded orbit that is not periodic,  $\lambda > 0$  is the standard numerical indicator of chaos, and that is how it is used here.

On a torus of an integrable system, in action–angle coordinates  $\varphi(t) = \varphi(0) + \nu(I)t$ , so a small change  $\delta I$  gives  $\delta\varphi(t) = \delta\varphi(0) + t(\partial\nu/\partial I)\delta I$ : nearby orbits separate *linearly*, by shear, and every  $\lambda$  is 0 (in any coordinates: near a compact torus all norms are comparable). In a chaotic region they separate exponentially (Figure 6.10). The limit is never reached numerically; what is computed is the finite-time exponent, which is how Figure 6.8 was made.

Finally, genericity. The course has claimed that integrability is the exception. Here is

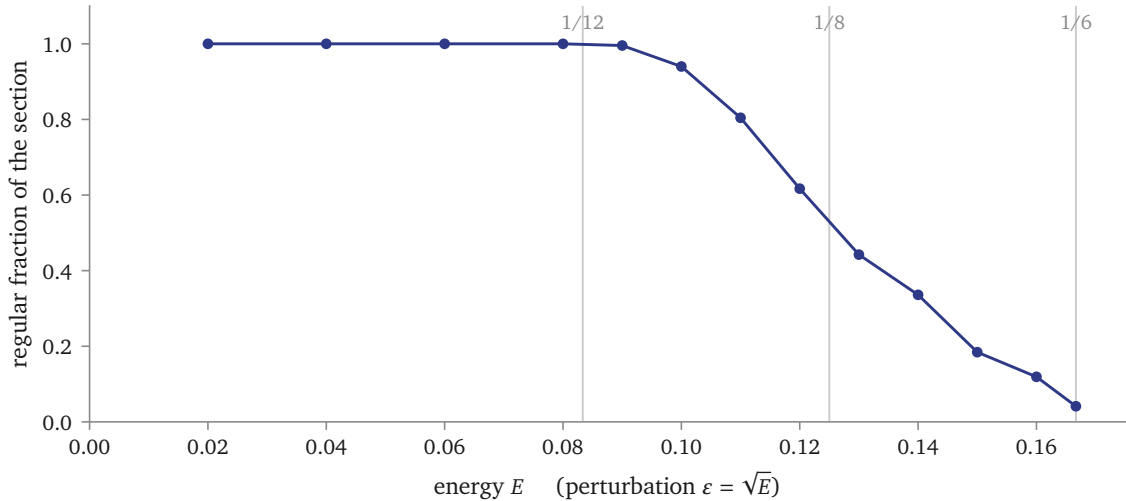


Figure 6.8: The regular fraction of the Hénon–Heiles section against the energy, i.e. against  $\varepsilon^2$ . A grid of launch points on the section is classified by a finite-time Lyapunov exponent (§ 6.7; regular if below 0.02 after  $t = 800$ ); a uniform grid estimates a fraction of area, the measure the section map preserves (Proposition 6.9). A measurement with two knobs, not a theorem (scripts/figures/surviving\_tori.py); Hénon and Heiles’s own estimate falls from 1 near  $E \approx 0.11$  (Toda, Fig. 1.5).

exactly what it can support.

1. In the perturbation problem (6.1), with  $H$  real-analytic in  $(\varphi, I, \varepsilon)$ ,  $H_0$  non-degenerate and the Poincaré set dense, there is no second integral, independent of  $H$ , that is a power series in  $\varepsilon$  with smooth coefficients (Theorem 6.3). What we proved (Proposition 6.5) is its first step: the leading term of any such integral is a function of  $H_0$ .
2. In the Hénon–Heiles family  $V_\lambda = \frac{1}{2}(x^2 + y^2) + x^2y + \frac{\lambda}{3}y^3$ ,  $\lambda \neq 0$  (Hénon–Heiles is  $\lambda = -1$ ), the known integrable members are  $\lambda = 1$ , which separates in the coordinates  $(y \pm x)/\sqrt{2}$ , and  $\lambda = 6$ , which has a second integral that is a polynomial of degree four, quadratic in the momenta.<sup>7</sup> For every other  $\lambda$  there is no second integral that is meromorphic in the complexified coordinates and momenta — in particular no rational or polynomial one (cited: Ito 1985, completed by Morales-Ruiz, Ramis and Simó 2007). Integrals that are only smooth, or defined on part of phase space, are not excluded. So the statement is: within this family, and for integrals of this class, the integrable members are isolated points of the parameter line.

That is all. No measure is put on “the space of all Hamiltonians”: that would need a space, a measure and a theorem, and the course has none of the three. What the two statements show is that integrability is not the default, and that it is not an accident of approximation — Figure 6.9 shows it destroyed by one, and item 2, at  $\lambda = -1$ , proves that meromorphic integrability is lost.

### Summary.

- The problem:  $H = H_0(I) + \varepsilon H_1(\varphi, I)$  (6.1). Poincaré’s theorem (Theorem 6.3, cited): for

<sup>7</sup>scripts/check-l6-algebra.py, check 5, which prints it.

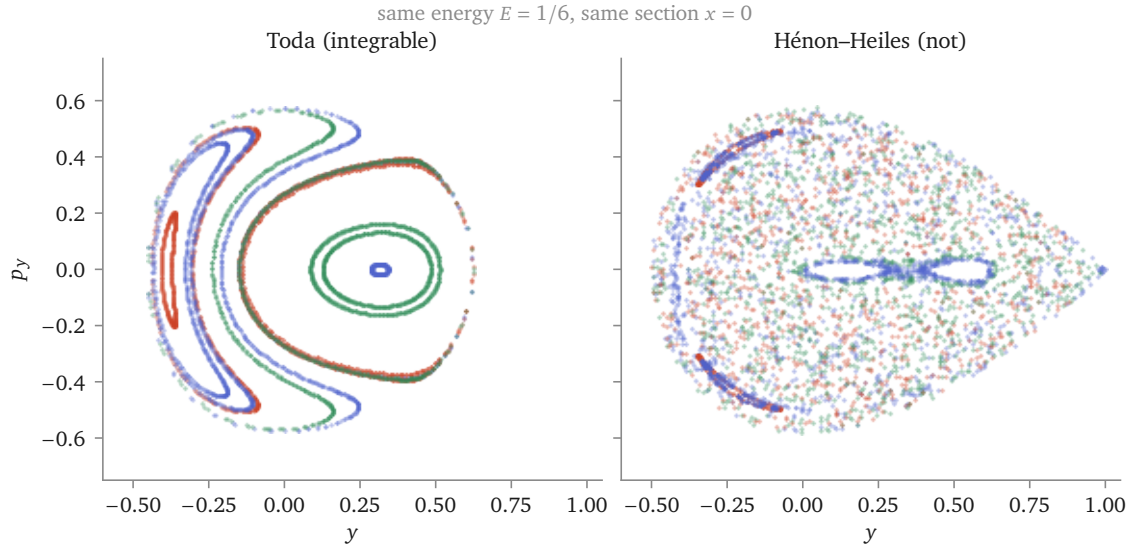


Figure 6.9: The same section  $x = 0$ ,  $p_x > 0$ , at the same energy  $E = 1/6$ , for the Toda chain (6.8) (left) and its cubic truncation Hénon–Heiles (right). Every plotted Toda orbit lies on a smooth curve; the truncation is mostly chaotic (scripts/figures/toda\_vs\_henon\_heiles.py).

$H_0$  non-degenerate and a dense Poincaré set (Definition 6.2) there is no second integral that is a power series in  $\varepsilon$ .

- *Small denominators*: first-order perturbation theory gives  $s_k = i h_k / (k \cdot \nu)$  (Proposition 6.4), singular on the dense set of resonances; the same resonances force any integral's leading term to depend on  $H_0$  alone (Proposition 6.5, proved at first order).
- *KAM* (Theorem 6.8, cited): for analytic, non-degenerate  $H$ , most tori survive, deformed; they carry Diophantine frequencies (Definition 6.6), which are almost all (Lemma 6.7). For  $n = 2$  they trap the orbits between them.
- *Poincaré–Birkhoff* (Theorem 6.10, cited): the section map preserves area (Proposition 6.9); a resonant circle breaks into an island chain of elliptic and hyperbolic periodic points, and the hyperbolic points' crossing curves form homoclinic tangles, the seed of chaos.
- *Hénon–Heiles* (6.7): regular  $\rightarrow$  mixed  $\rightarrow$  chaotic as  $E = \varepsilon^2$  grows. It is the cubic truncation of the periodic Toda chain (6.8), which stays regular at the same energy.
- *Lyapunov exponents* (Definition 6.11) vanish on tori. *Genericity*, and no more: Poincaré's theorem, and the isolation of the integrable members of the Hénon–Heiles family.

**Tutorial.** Exercise Sheet 6.

## References

Arnold, *Mathematical Methods of Classical Mechanics*, App. 8 (the perturbation problem, small denominators, Kolmogorov's theorem with its hypotheses, §§A–D) and App. 9 (Poincaré's geometric theorem); Kozlov, *Integrability and non-integrability in Hamiltonian mechanics*, Russian Math. Surveys 38:1 (1983), ch. IV §1 (Poincaré's theorem, Theorems 1–2); Ott, *Chaos in Dynamical Systems*, 2nd ed., §7.2 (KAM, the Poincaré–Birkhoff alternation), §7.3

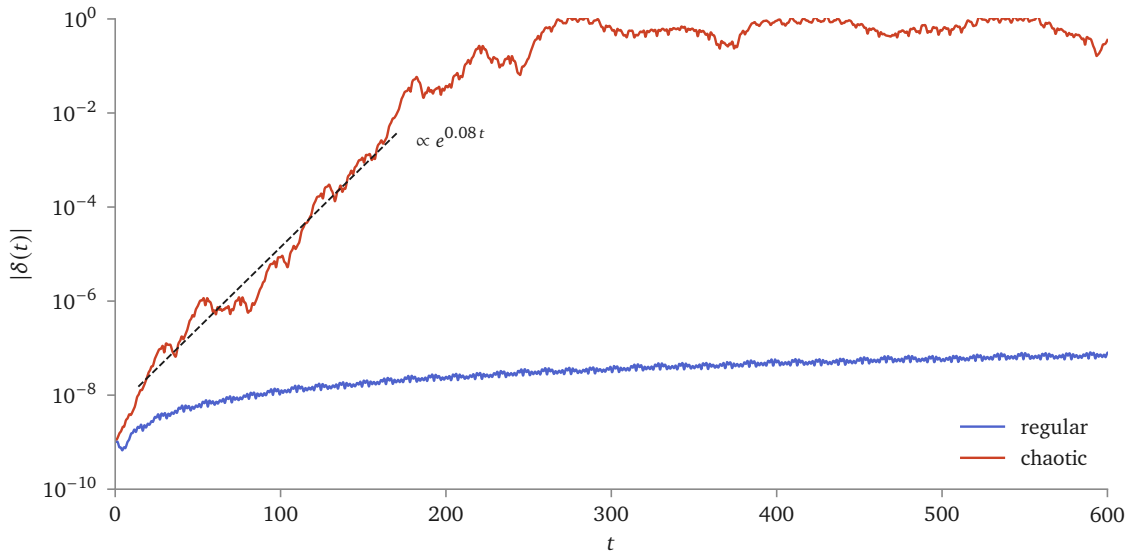


Figure 6.10: Two pairs of Hénon–Heiles orbits at  $E = 1/8$ , each started  $10^{-9}$  apart. Launched inside an island (blue), the pair separates linearly. Launched in the chaotic sea (red), it separates like  $e^{\lambda t}$  until the distance reaches the size of the region (scripts/figures/lyapunov\_separation.py).

(the standard map), §4.4 (Lyapunov exponents); Toda, *Theory of Nonlinear Lattices*, §1.2 (Hénon–Heiles) and §2.9 (the three-particle chain and its reduction); Hénon and Heiles, *Astron. J.* 69 (1964) 73; H. Ito, *Kodai Math. J.* 8 (1985) 120, and J.J. Morales-Ruiz, J.-P. Ramis and C. Simó, *Ann. Sci. ENS* 40 (2007) 845, §6 (the Hénon–Heiles family). **Conventions.** Arnold’s App. 9 writes the symplectic form as  $dI \wedge d\varphi$ , the negative of ours; areas and fixed points do not see the sign.